

HAT-P-36b

R-0837

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250211 · created 2026-07-28T18:34:46+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460717.9517 +/- 0.0062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.046
Transit depth (Rp/Rs)^2	2.43 +/- 1.42 [%]
Orbital Inclination (inc)	81.07 +/- 2.77
Airmass coefficient 1 (a1)	5.16 +/- 3.28
Airmass coefficient 2 (a2)	-1.64 +/- 1.41
Scatter in the residuals of the lightcurve fit is	67.37 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	3.0
Transit Duration (day)	0.062 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.046 vs 0.126 (deviation 0.63σ)
Duration fit vs published	0.062 d vs 0.0925 d (deviation 2.1σ)
Mid-time O–C	13.5 min
Depth SNR	3.3
Residual scatter	67.37 %

Light curve

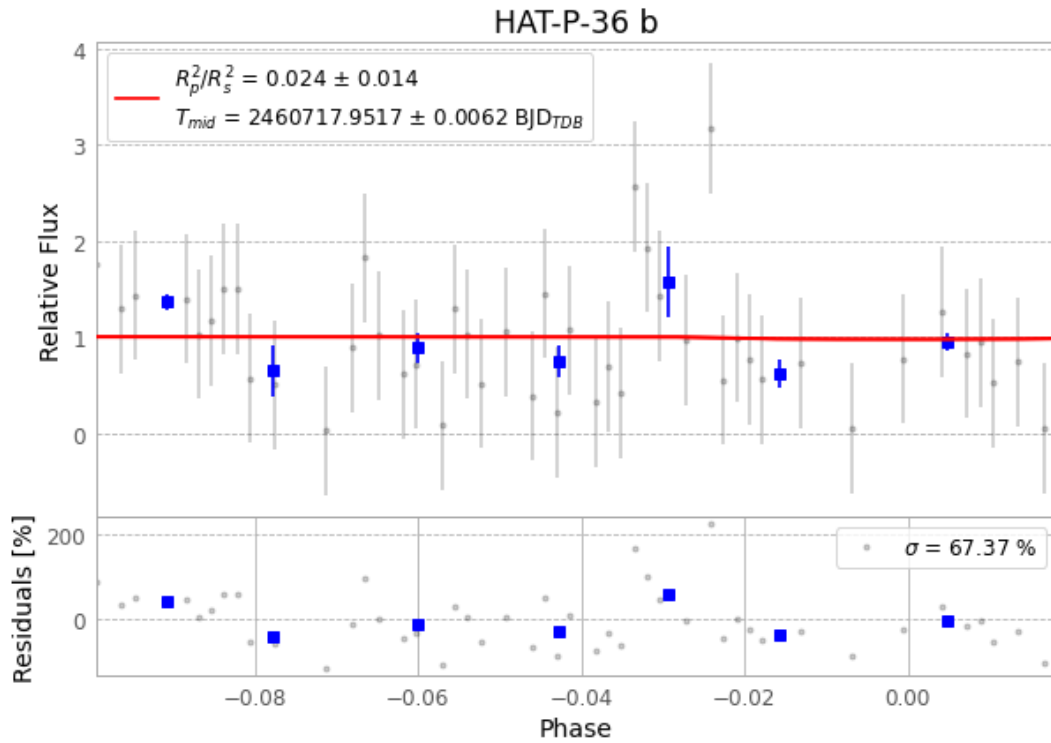


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-02-11.png (SHA-256 2f285830fc770929...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 454], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cb0f692afc816adf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), HATP-36250211073016.FITS → HATP-36250211111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250211.log (2a89c55c9509...), FinalLightCurve_HAT-P-36 b_2025-02-11.png (2f285830fc77...), FinalLightCurve_HAT-P-36 b_2025-02-11.csv (00a577e945f6...)

Compute resources

Wall time 140.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 18:34Z.